First year Aerospace Engineering

Mathematics - Geometry

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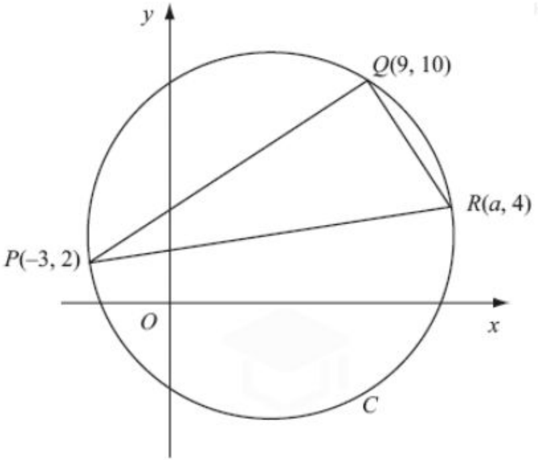
Geometry 1



The points $P(-3, 2)$, $Q(9, 10)$ and $R(a, 4)$ lie on the circle C, as shown in the diagram above.

Given that PR is a diameter of C,

(a) show that $a = 13$,

(3)

(b) find an equation for C.

(5)

(Total 8 marks)

Geometry 2

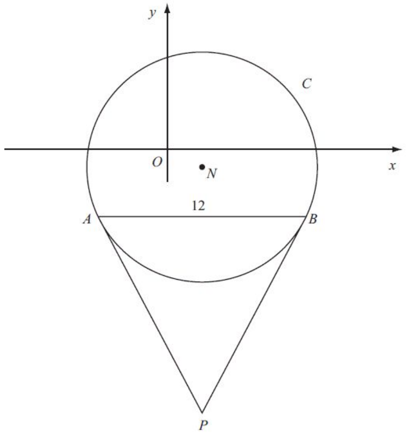


Figure 3

Figure 3 shows a sketch of the circle C with centre N and equation

$$(x-2)^2 + (y+1)^2 = \frac{169}{4}$$

(a) Write down the coordinates of N.
(2)

(b) Find the radius of C.
(1)

The chord AB of C is parallel to the x-axis, lies below the x-axis and is of length 12 units as shown in Figure 3.

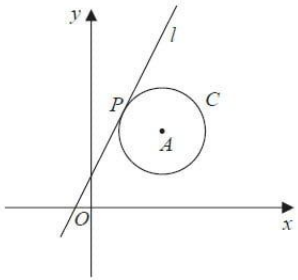
(c) Find the coordinates of A and the coordinates of B.
(5)

(d) Show that angle $ANB = 134.8°$, to the nearest 0.1 of a degree.
(2)

The tangents to C at the points A and B meet at the point P.

(e) Find the length AP, giving your answer to 3 significant figures.
(2)

Geometry 3



Not to scale

Figure 3

The circle C has centre A with coordinates $(7, 5)$.

The line l, with equation $y = 2x + 1$, is the tangent to C at the point P, as shown in Figure 3.

(a) Show that an equation of the line PA is $2y + x = 17$

(3)

(b) Find an equation for C.

(4)

The line with equation $y = 2x + k$, $k \neq 1$ is also a tangent to C.

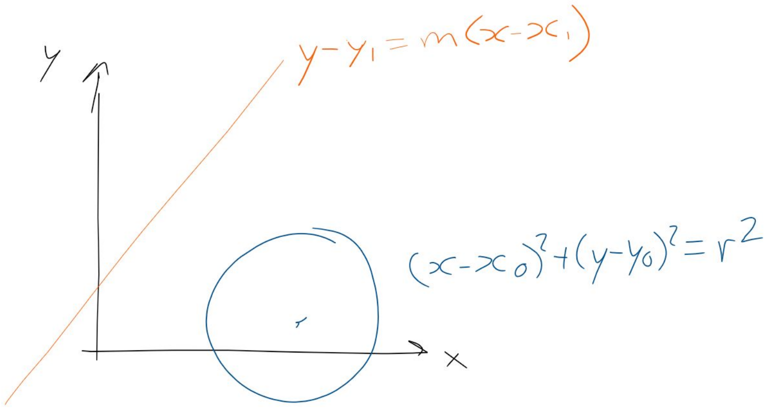
(c) Find the value of the constant k.

(3)

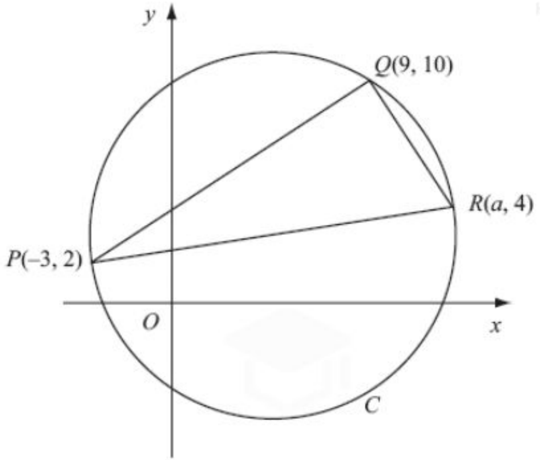
Geometry 4

Let $y_1=2$, $x_1=1$, $m=1$
The blue circle has centre (4,1) and a diameter of 4 units

a) Find the shortest distance between the orange and blue line
b) Find the equation of the line that connects the blue and orange line through the shortest distance between them.
c) Let point A be the intersection between this new line and the orange line. What are the coordinates of point A?
d) What are the equations of the lines that pass through A and are tangent to the blue line?



Geometry 1



The points $P(-3, 2)$, $Q(9, 10)$ and $R(a, 4)$ lie on the circle C, as shown in the diagram above.

Given that PR is a diameter of C,

(a) show that $a = 13$, Pythagorean thm

(3)

(b) find an equation for C.

(5)

(Total 8 marks)

$(x-5)^2+(y-3)^2=65$

Geometry 2

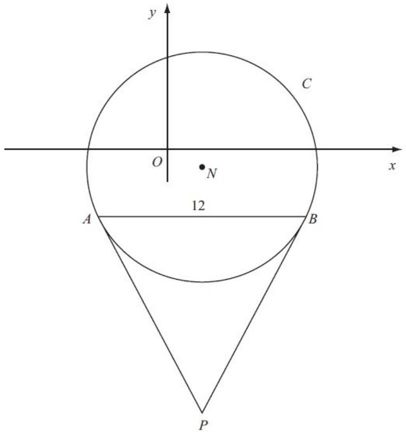


Figure 3

Figure 3 shows a sketch of the circle C with centre N and equation

$$(x-2)^2 + (y+1)^2 = \frac{169}{4}$$

(a) Write down the coordinates of N. 2, -1 **(2)**

(b) Find the radius of C. 13/2 **(1)**

The chord AB of C is parallel to the x-axis, lies below the x-axis and is of length 12 units as shown in Figure 3.

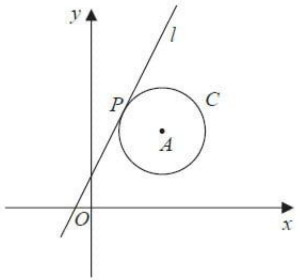
(c) Find the coordinates of A and the coordinates of B. A: (-4, -3.5), B: (8, -3.5) **(5)**

(d) Show that angle $ANB = 134.8°$, to the nearest 0.1 of a degree. theta/2 = arctan(6/2.5) **(2)**

The tangents to C at the points A and B meet at the point P.

(e) Find the length AP, giving your answer to 3 significant figures. 15.6 **(2)**

Geometry 3



Not to scale

Figure 3

The circle C has centre A with coordinates $(7, 5)$.

The line l, with equation $y = 2x + 1$, is the tangent to C at the point P, as shown in Figure 3.

(a) Show that an equation of the line PA is $2y + x = 17$ PA goes through A and has gradient -½

(3)

(b) Find an equation for C. $(x-7)^2+(y-5)^2=20$

(4)

The line with equation $y = 2x + k$, $k \neq 1$ is also a tangent to C.

(c) Find the value of the constant k. $k=-19$

(3)

Geometry 4

Let y_1=2, x_1=1, m=1
The blue circle has centre (4,1) and a diameter of 4 units

a) Find the shortest distance between the orange and blue line
0.8284

b) Find the equation of the line that connects the blue and orange line through the shortest distance between them. y=-x+5

c) Let point A be the intersection between this new line and the orange line. What are the coordinates of point A? (2,3)

d) What are the equations of the lines that pass through A and are tangent to the blue line?
y=3, x=2